

L2 Angstrom Audit Summary

Prepared by Cyfrin

Version 2.1

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Contents

1 About Cyfrin 2

2 About This Document 2

3 Disclaimer 2

4 Copyright and License 2

5 Audit Overview 3

6 Audit Scope 3

7 Risk Classification 4

8 Protocol Summary 4

9 Attestation 4

1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Sorella Labs on L2 Angstrom. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.1 dated October 1, 2025, was delivered to Sorella Labs, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 10 days, the Cyfrin team conducted an audit on the L2 Angstrom code provided by Sorella Labs. The findings consist of 1 High, 4 Medium & 5 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	L2 Angstrom
Repository	l2-angstrom
Commit	386baff9f903...
Audit Timeline	Sept 9th - Sept 22nd, 2025
Methods	Manual Review, Stateless Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	1	1	0
Medium Risk	4	4	0
Low Risk	5	3	2
Informational	8	5	3
Gas Optimizations	6	3	3
Total	24	16	8

6 Audit Scope

The audit scope was limited to:

```
src/libraries/CompensationPriceFinder.sol
src/libraries/Math512Lib.sol
src/libraries/MixedSignLib.sol
src/libraries/PoolKeyHelperLib.sol
src/libraries/Q96MathLib.sol
src/libraries/TickIterator.sol
src/libraries/TickLib.sol
src/modules/UniConsumer.sol
src/types/PoolRewards.sol
src/AngstromL2.sol
src/hook-config.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

The L2 deployment of Angstrom to priority-ordered chains is a Uniswap V4 hook that imposes a tax on just-in-time (JIT) liquidity provision and top-of-block swaps. MEV is redistributed to liquidity providers (LPs) based on an effective swap price that prioritises rewarding LPs who are more heavily affected by the arbitrage that is assumed to occur in the top-of-block swap. The assumption is that the entity that wins the bid will take advantage of some discrepancy in pricing between the AMM and other (likely centralized) trading venues, so LPs further below the effective price suffer greater loss-versus-rebalancing (LVR) and should be more heavily compensated.

9 Attestation

Cyfrin attests that it performed a security review of the L2 Angstrom code provided by Sorella Labs.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-10-01-cyfrin-sorella-l2-angstrom-v2.1.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Sorella Labs has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=4d990f46fe55db6bff491b74a2aba6a00f7e8c917e54be91ef58c3071bb8e7bf
F=2025-10-01-cyfrin-sorella-l2-angstrom-v2.1-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.